New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

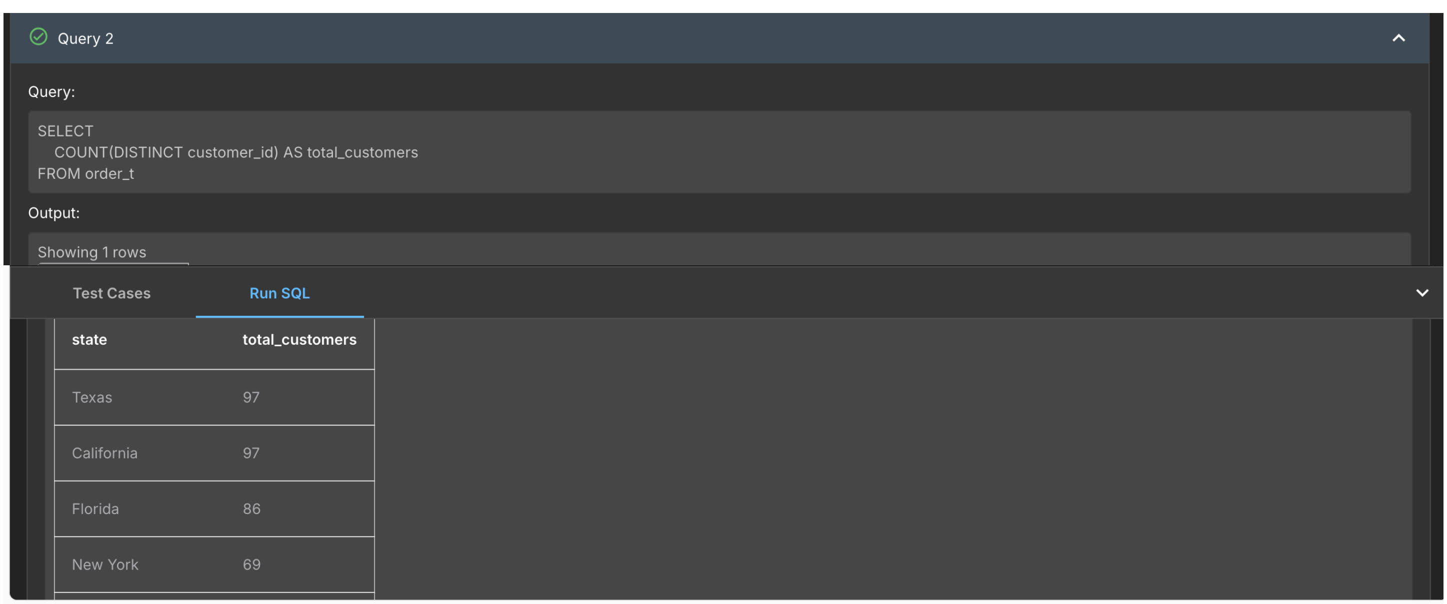
Solution Query:

```
SELECT
    c.state,
    COUNT(DISTINCT c.customer_id) AS total_customers
FROM customer_t c
JOIN order_t o
    ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY total_customers DESC;
SELECT
    COUNT(DISTINCT customer_id) AS total_customers
FROM order_t;
```

Example:-

```
select *
from online_customer
where CUSTOMER_GENDER = 'F';
```

Output:


Query 2

Query:

```
SELECT
    COUNT(DISTINCT customer_id) AS total_customers
FROM order_t
```

Output:

Showing 1 rows

Test Cases   Run SQL

state	total_customers
Texas	97
California	97
Florida	86
New York	69

<Attach the screenshot of your output table>

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

Result: Passed

Query 1

Query:

```
SELECT
    c.state,
    COUNT(DISTINCT c.customer_id) AS total_customers
FROM customer_t c
JOIN order_t o
    ON c.customer_id = o.customer_id
```

- A sample screenshot is provided below.

SQL queries passed

Query 1 (Passed): select * from online_customer where CUSTOMER_GENDER = "F"
Output:
Showing first 10 rows out of 24 rows

CUSTOMER_ID	CUSTOMER_FNAME	CUSTOMER_LNAME	CUSTOMER_EMAIL	CUSTOMER_PHONE	ADDRESS_ID	CUSTOMER_CREATIO...	CUSTOMER_USERNAME
1	Jennifer	Wilson	jen_w@gmail.com	9776363306	909	1991-06-01	jen_w
3	Komal	Choudhary	ch_komal@yahoo.co.IN	9178234526	911	2002-06-26	komalc
5	Ramya	Ravinder	ramya_r23@gmail.com	7732341567	913	2006-02-12	rramya
6	Anita	Goswami	agoswami@gmail.com	9873245623	914	2006-03-13	anitag
7	Ashwathi	Bhatt	ash_bhat@yahoo.co.IN	9773636307	915	2007-04-15	abhatt

Observations and Insights:

- **There are 994 total customers**
- Most of the customers are located in Texas and California
- The least amount of customers are located in Arizona

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

SELECT

p.vehicle_maker,

COUNT(*) AS total_orders

FROM order_t o

JOIN product_t p

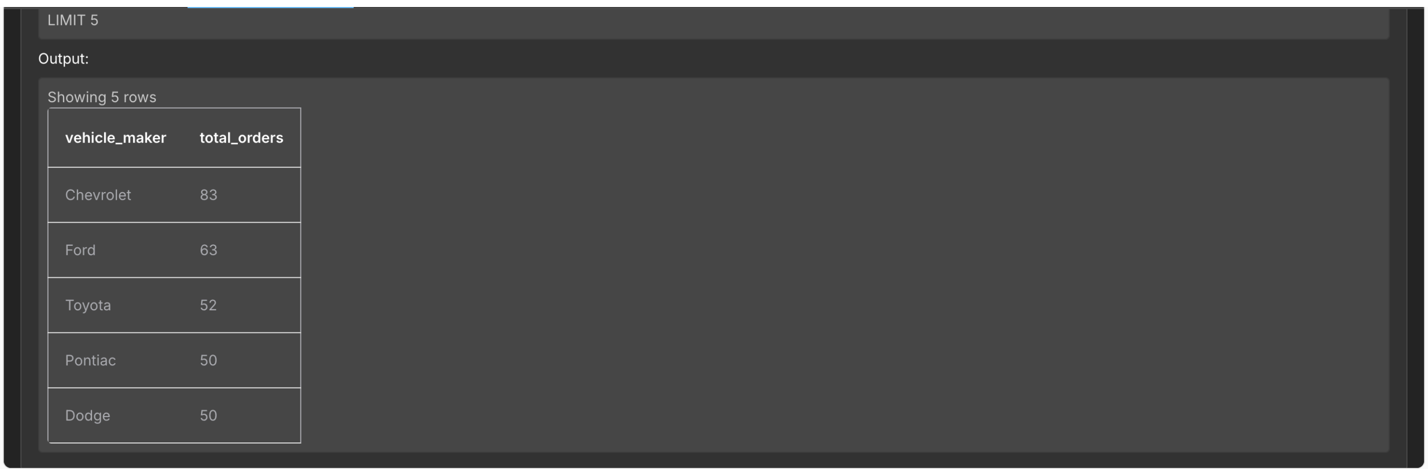
    ON o.product_id = p.product_id

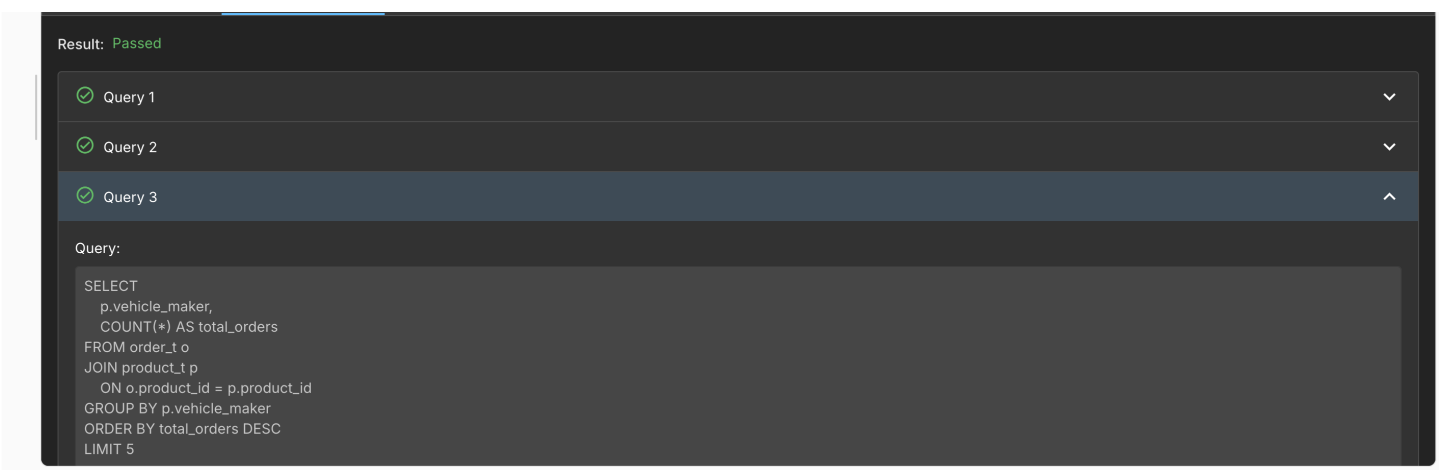
GROUP BY p.vehicle_maker

ORDER BY total_orders DESC

LIMIT 5;

Output:





- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

Observations and Insights:

- **Chevrolet is the most preferred vehicle maker**
- Dodge is the least preferred vehicle maker
-

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, total_orders
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(*) AS total_orders,
        ROW_NUMBER() OVER (
            PARTITION BY c.state
            ORDER BY COUNT(*) DESC
        ) AS rn
    FROM customer_t c
    JOIN order_t o
        ON c.customer_id = o.customer_id
    JOIN product_t p
        ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) t
WHERE rn = 1
ORDER BY state;
```

Output:

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

-

Query 4

Query:

```
SELECT state, vehicle_maker, total_orders
FROM (
  SELECT
    c.state,
    p.vehicle_maker,
    COUNT(*) AS total_orders,
    ROW_NUMBER() OVER (
      PARTITION BY c.state
      ORDER BY COUNT(*) DESC
    ) AS rn
  FROM customer_t c
  JOIN order_t o
    ON c.customer_id = o.customer_id
  JOIN product_t p
    ON o.product_id = p.product_id
  GROUP BY c.state, p.vehicle_maker
) t
WHERE rn = 1
ORDER BY state
```

Output:

Showing first 10 rows out of 49 rows

Output:

Showing first 10 rows out of 49 rows

state	vehicle_maker	total_orders
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arkansas	Volkswagen	1
California	Nissan	6
Colorado	Chevrolet	5
Connecticut	Volvo	2
Delaware	Mitsubishi	2
District of Columbia	Chevrolet	4
Florida	Toyota	7

Observations and Insights:

- Some vehicle makers are popular in multiple states, while other seem to be limited by regional popularity
- Maybe focus on marketing and having stock of regionally popular brands to the appropriate region in order to maximize sales

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT
    quarter_number,
    AVG(rating) AS avg_rating_per_quarter
FROM (
    SELECT
        quarter_number,
        CASE
            WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5
        END AS rating
    FROM order_t
) t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Query 7

Query:

```
SELECT
   quarter_number,
   AVG(rating) AS avg_rating_per_quarter
FROM (
   SELECT
      quarter_number,
      CASE
         WHEN customer_feedback = 'Very Bad' THEN 1
         WHEN customer_feedback = 'Bad' THEN 2
         WHEN customer_feedback = 'Okay' THEN 3
         WHEN customer_feedback = 'Good' THEN 4
         WHEN customer_feedback = 'Very Good' THEN 5
      END AS rating
   FROM order_t
) t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_rating_per_quarter

Output:

Showing 4 rows

quarter_number	avg_rating_per_quarter
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

Observations and Insights:

- **We see a decline in the average rating as the quarters go by. It goes from around 3.5 in Q1 to around 2.4 in Q4.**
- Dissatisfaction seems to be rising. What is the element causing it? Once we identify that, we can move forward.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT

    quarter_number,


    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS very_bad_pct,


    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
bad_pct,


    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
okay_pct,


    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
good_pct,


    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(*),
2) AS very_good_pct


FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number;
```

Output:

Output:

Showing 4 rows

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Query 5

Query:

```
SELECT
   quarter_number,

   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS very_bad_pct,

   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS bad_pct,

   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2) AS okay_pct,

   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS good_pct,

   ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS very_good_pct

FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

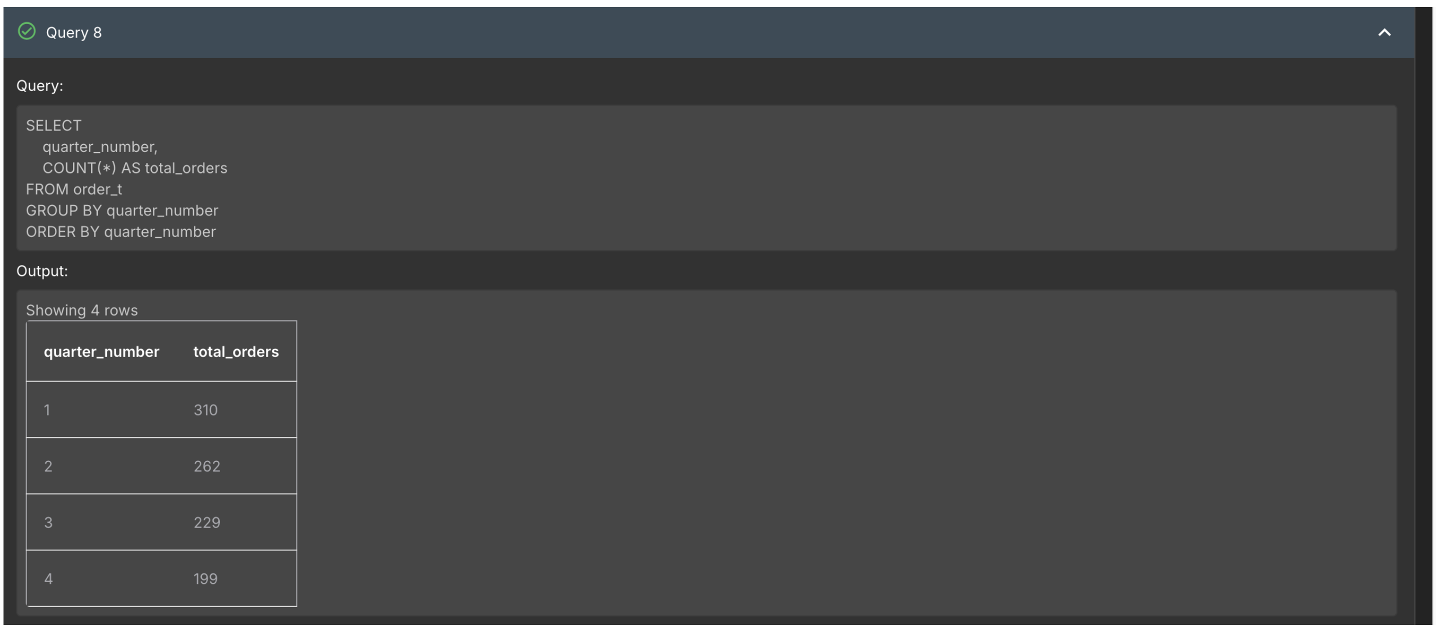
Observations and Insights:

- **The customers seem to be more dissatisfied as the quarters go by. For example, very bad feedback is around 11% in Q1 while it is around 31% in Q4.**
- We should focus on investigating what is causing customers to become more dissatisfied. Customer service? Selection? Shipping time? Once that cause is identified we can improve feedback.

Question 6: What is the trend of the number of orders by quarter?

```
SELECT
    quarter_number,
    COUNT(*) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Query 8

Query:

```
SELECT
    quarter_number,
    COUNT(*) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

Observations and Insights:

- **Orders are decreasing along with customer satisfaction. This is likely correlated with the decreasing sales. We need to identify what the cause is of this decline in sales.**

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
    quarter_number,
    net_revenue,
    prev_revenue,
    ((net_revenue - prev_revenue) * 100.0 / prev_revenue) AS qoq_percentage_change
FROM (
    SELECT
        quarter_number,
        net_revenue,
        LAG(net_revenue) OVER (ORDER BY quarter_number) AS prev_revenue
    FROM (
        SELECT
            quarter_number,
            SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
        FROM order_t
        GROUP BY quarter_number
    )
)
ORDER BY quarter_number;
```

Output:

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

Query 9

Query:

```
SELECT
   quarter_number,
   net_revenue,
   prev_revenue,
   ((net_revenue - prev_revenue) * 100.0 / prev_revenue) AS qoq_percentage_change
FROM (
   SELECT
      quarter_number,
      net_revenue,
      LAG(net_revenue) OVER (ORDER BY quarter_number) AS prev_revenue
   FROM (
      SELECT
         quarter_number,
         SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
      FROM order_t
      GROUP BY quarter_number
   )
)
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	prev_revenue	qoq_percentage_change
1	18032549.899600018		

Output:

Showing 4 rows

quarter_number	net_revenue	prev_revenue	qoq_percentage_change
1	18032549.899600018		
2	13122995.7562	18032549.899600018	-27.226067143776024
3	8882298.8449	13122995.7562	-32.31500634522776
4	8573149.280599998	8882298.8449	-3.480512980910439

Observations & Insights:

- There is a consistent decline in revenue, showing weakening sales over time.
- However, the decline seems to slow down from Q3 to Q4, which means it may be stabilizing.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(*) AS total_orders,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Query 10

Query:

```
SELECT
    quarter_number,
    COUNT(*) AS total_orders,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	total_orders	net_revenue
1	310	18032549.899600018
2	262	13122995.7562
3	229	8882298.8449
4	199	8573149.280599998

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

Observations and Insights:

- **Again we see declining revenue and total orders. We need to identify the element that makes people order less and be more dissatisfied over time.**

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    AVG(o.discount) AS avg_discount
FROM customer_t c
JOIN order_t o
    ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount DESC;
```

Note: The '`discount`' field in the '`order_t`' table is stored as a percentage, i.e. `0.6` represents a discount of `0.6%`, not `60%`.

Output:

Query 11

Query:

```
SELECT
    c.credit_card_type,
    AVG(o.discount) AS avg_discount
FROM customer_t c
JOIN order_t o
    ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	avg_discount
laser	0.643846153846154
mastercard	0.6294999999999998
maestro	0.6242187499999999
visa-electron	0.623469387755102
china-unionpay	0.6221739130434784

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the "SQL Query Passed" along with the output table.

Observations and Insights:

- **Customers using laser cards have the highest discount, so they get the best pricing.**
- There is little variation over the other credit cards, suggesting relatively similar pricing strategies.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

SELECT

quarter_number,

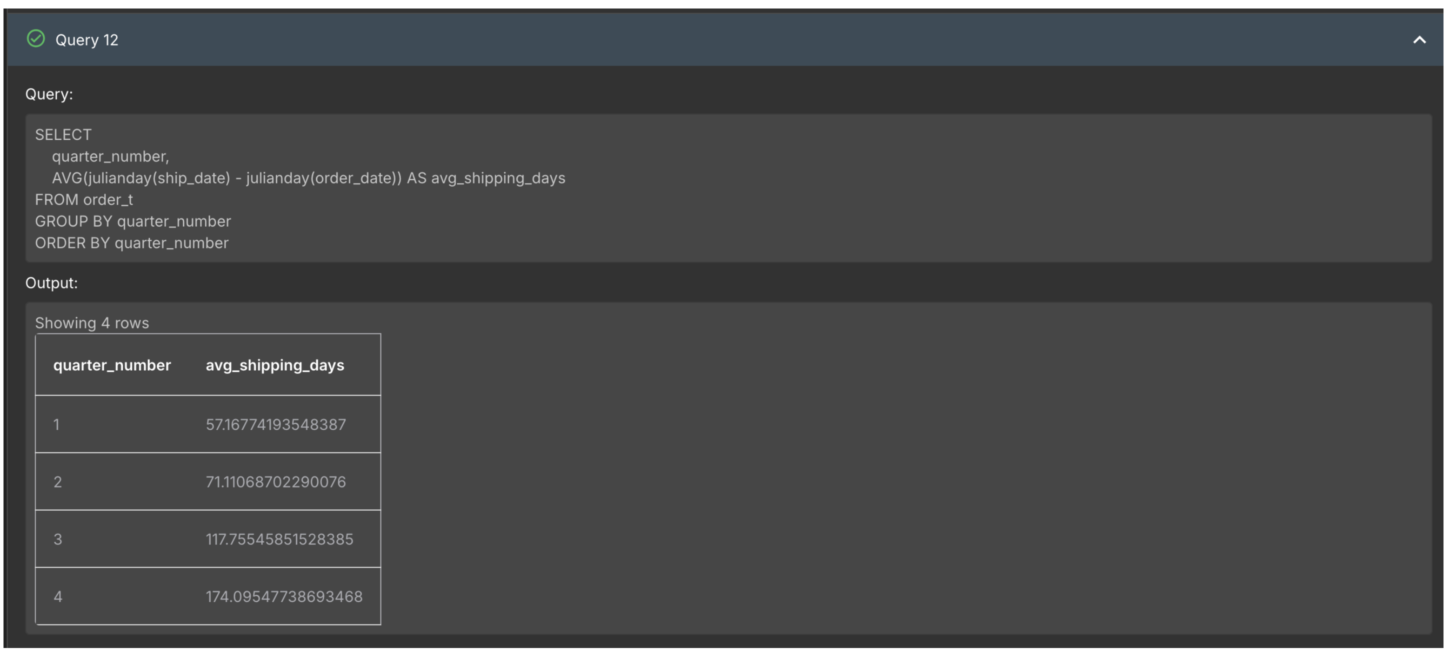
AVG(julianday(ship_date) - julianday(order_date)) AS avg_shipping_days

FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number;

Output:


Query 12

Query:

```
SELECT
    quarter_number,
    AVG(julianday(ship_date) - julianday(order_date)) AS avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_shipping_days
1	57.16774193548387
2	71.11068702290076
3	117.75545851528385
4	174.09547738693468

- The screenshot should display 5-10 rows of the output
- Make sure the screenshot contains the “SQL Query Passed” along with the output table.

Observations and Insights:

- **The shipping time has more than doubled since Q1, which could indicate a factor in customer dissatisfaction.**

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48610993.7813	1000	994	3.135
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
8573149.2806	199	97.964	21.5

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- **Rising negative feedback and declining ratings suggest issues in service or delivery. Focus on fixing shipping delays and after-sales support.**
- Increasing shipping time is likely driving dissatisfaction, so streamline delivery processes to improve speed and reliability.
- Focus on high-demand regions and popular vehicle makers while reviewing discount strategies to boost sales and retain customers.